

Chapter 10 Biomolecules

For **NEET UG**

- NCERT Based
- High Yield Notes

INTRODUCTION

Biomolecules are the natural compounds present in living organisms. They are essential for structure, function, growth, metabolism and maintenance of life. They are mainly of four types -

- ★ Carbohydrates
- ★ Proteins
- ★ Lipids
- ★ Nucleic Acids

IMPORTANCE OF BIOMOLECULES

- Builds and repairs tissues
- Provides energy
- Regulates metabolic processes
- Stores genetic information
- Protects from diseases

ELEMENTS COMMONLY PRESENT

C	H	O	N	P	S
↓	↓	↓	↓	↓	↓
Carbon (Backbone of life)	Hydrogen (Most abundant elements)	Oxygen	Nitrogen (Proteins & Nucleic acids)	Phosphorus Sulphur (Nucleic acids & Proteins)	

CLASSIFICATION Based on Nature

- ① Primary Biomolecules → Carbohydrates, Proteins, Nucleic Acids
- ② Secondary Biomolecules → Lipids, Vitamins, Hormones, Enzymes

1. CARBOHYDRATES

- Most abundant biomolecules.
- Composed of C, H, O (in 1:2:1 ratio)
- General formula: $(CH_2O)_n$
- Are polyhydroxy aldehydes or ketones or compounds which yield them on hydrolysis.

CLASSIFICATION

Monosaccharides
(One sugar unit)
e.g., Glucose, Fructose

Oligosaccharides
(2-10 sugar units)
e.g., Sucrose, Maltose

Polysaccharides
(Many sugar units)
e.g., Starch, Cellulose, Glycogen

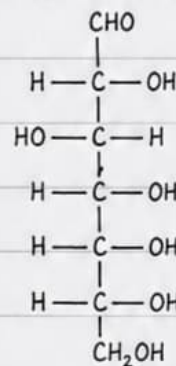


Remember:

Monosaccharides are sweet & soluble in water while polysaccharides are generally insoluble and non-sweet.

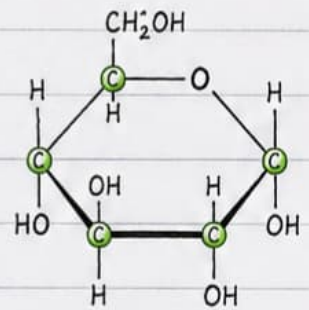
STRUCTURE OF GLUCOSE

Open chain form



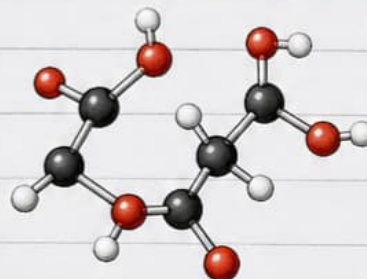
(D-Glucose)

Cyclic form



(α -D-Glucopyranose)

3D MODEL OF GLUCOSE



● C
● O
○ H

(Ball & Stick Model)



Biomolecules are the **CHEMICAL BASIS OF LIFE**.

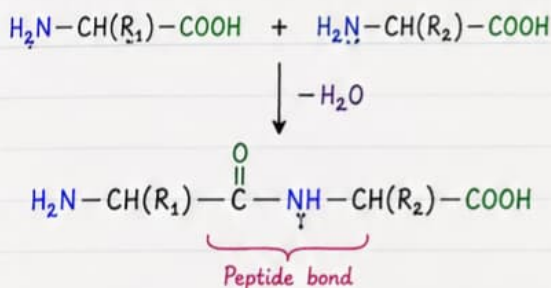
Understand them, respect life. ♥

Stay Focused
Do Your Best
Crack NEET

2. PROTEINS

- Proteins are the most abundant organic biomolecules after water.
- They are polymers of α -amino acids linked by peptide bonds.
- Contain C, H, O, N and sometimes S (in cysteine & methionine).
- Perform structural, catalytic, hormonal, transport, immune and storage functions.
- On hydrolysis $\rightarrow$ give α -amino acids.

PEPTIDE BOND FORMATION

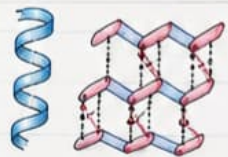


LEVELS OF PROTEIN STRUCTURE



① Primary Structure

Linear sequence of amino acids linked by peptide bonds.



② Secondary Structure

Local folding of the polypeptide chain by H-bonding.

- α -Helix $\rightarrow$ Right handed spiral
- β -Pleated sheet $\rightarrow$ Extended sheet



③ Tertiary Structure

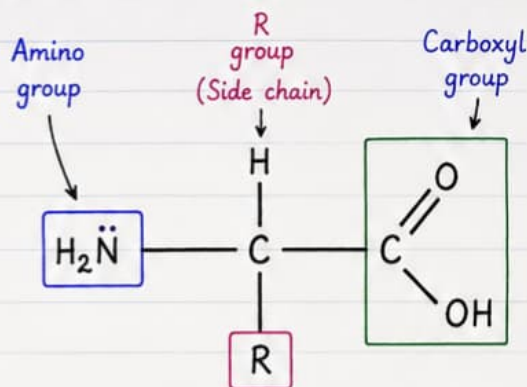
Overall 3D shape of a polypeptide chain due to R-group interactions (disulfide bonds, ionic, H-bonds, van der Waals forces, hydrophobic interactions).



④ Quaternary Structure

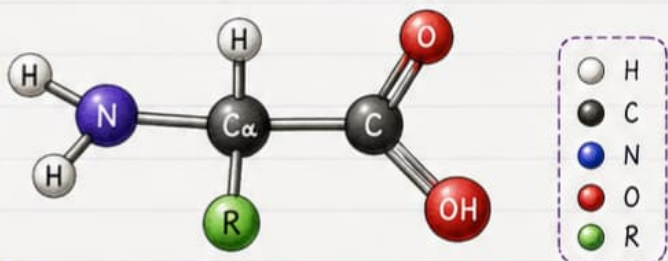
Association of two or more polypeptide chains (subunits) to form of functional protein.

AMINO ACIDS - The Building Blocks



- General formula: $\text{H}_2\text{N}-\text{CH}(\text{R})-\text{COOH}$
- Except glycine ($\text{R} = \text{H}$), all α -amino acids are optically active.

3D MODEL OF AN α -AMINO ACID (General Structure)



FUNCTIONS OF PROTEINS (Examples)

- ★ **Structural** - Collagen, Keratin
- ★ **Catalytic (Enzymes)** - Amylase, Pepsin
- ★ **Transport** - Hemoglobin (transports O_2)
- ★ **Hormonal** - Insulin, Growth hormone
- ★ **Defensive** - Antibodies (Immunoglobulins)
- ★ **Storage** - Ferritin (stores iron), Casein (milk protein)

Quick Revision

- Proteins are polymers of amino acids.
- Peptide bond links amino acids.
- Proteins have 4 levels of structure.
- Structure decides the function.
- Denaturation $\rightarrow$ Loss of 3D structure (by heat, pH, etc.) $\rightarrow$ Loss of function.

First learn the basics, then master the details!

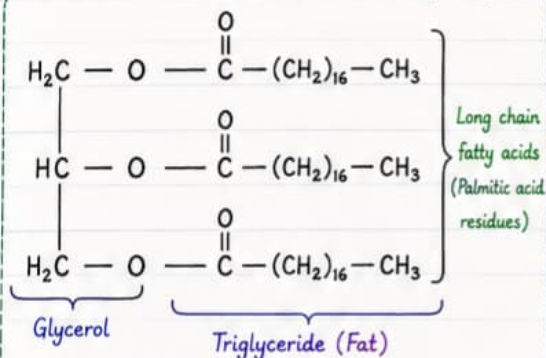
3. LIPIDS

- Lipids are a heterogeneous group of biomolecules insoluble in water but soluble in organic solvents.
- They are esters of fatty acids with alcohols.
- Do not form polymers.
- Act as energy store, membrane components, hormones & vitamins.

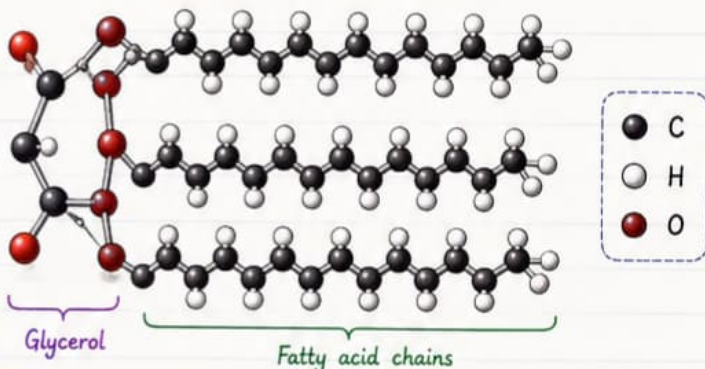
⇒ TYPES OF LIPIDS ⇐

Type	Example	Function
Fats & Oils	Triglycerides	Energy storage, insulation, protection
Phospholipids	Lecithin	Major component of cell membranes
Steroids	Cholesterol, Testosterone, Estrogen	Hormones, membrane stability
Waxes	Beeswax	Waterproofing, protection

STRUCTURE OF TRIGLYCERIDE (FAT)



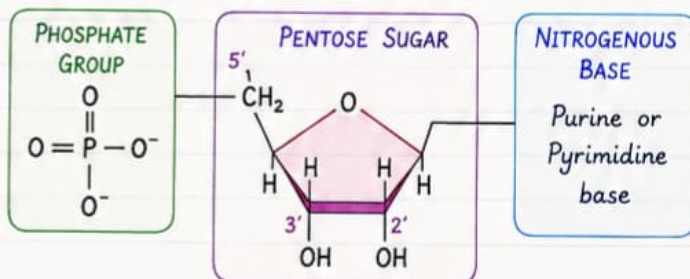
3D MODEL OF A TRIGLYCERIDE



4. NUCLEIC ACIDS

- Nucleic acids are polymers of nucleotides.
- Two types - DNA (Deoxyribonucleic acid) and RNA (Ribonucleic acid)
- Store and transmit genetic information.
- Found in nucleus, mitochondria, chloroplasts, ribosomes and some viruses.

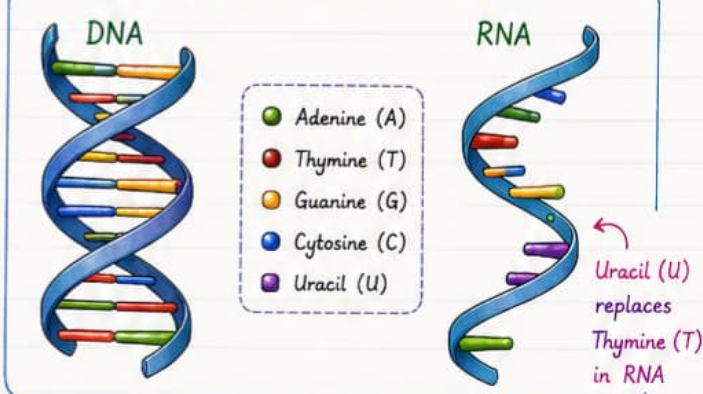
STRUCTURE OF A NUCLEOTIDE



DIFFERENCE BETWEEN DNA & RNA

Feature	DNA	RNA
Full form	Deoxyribonucleic acid	Ribonucleic acid
Sugar	2-Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Structure	Double stranded (Helical)	Single stranded
Location	Nucleus, Mitochondria, Chloroplasts	Nucleus, Cytoplasm, Ribosomes
Function	Storage of genetic information	Expression of genetic information (protein synthesis)

3D MODELS OF DNA & RNA



REMEMBER:

- Proteins → Work.
- Carbohydrates → Instant Energy.
- Fats → Stored Energy.
- Nucleic Acids → Information.
- Life is a beautiful combination of Biomolecules. ♥



QUICK REVISION

- ✓ Biomolecules are essential for life.
- ✓ Each biomolecule has a unique structure → specific function.
- ✓ Together they maintain life processes.



5. VITAMINS

Vitamins are **organic micronutrients** required in small amounts but essential for normal growth and metabolism. ☆

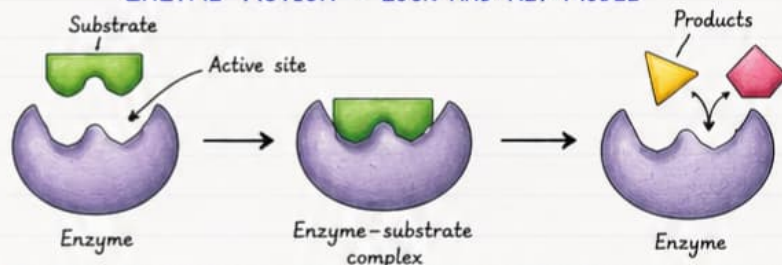
FAT SOLUBLE VITAMINS	Functions & Sources
Vitamin A	Good for vision, skin, immunity. Sources: Carrot, Milk, Egg, Green leafy vegetables.
Vitamin D	Helps in absorption of Ca^{2+} and P. Good for bones & teeth. Sources: Sunlight, Egg yolk, Fish liver oil.
Vitamin E	Antioxidant, protects cell membranes. Sources: Nuts, Seeds, Vegetable oils.
Vitamin K	Necessary for blood clotting. Sources: Green leafy vegetables, Soya, Tomato.

WATER SOLUBLE VITAMINS	Functions & Sources
Vitamin B ₁ (Thiamine)	Carbohydrate metabolism, healthy nerves. Sources: Whole grains, Pulses, Nuts.
Vitamin B ₂ (Riboflavin)	Healthy skin & eyes, energy release. Sources: Milk, Eggs, Green vegetables.
Vitamin B ₃ (Niacin)	Good for skin, nerves, digestion. Sources: Meat, Fish, Peanuts.
Vitamin B ₆ (Pyridoxine)	Amino acid metabolism, RBC formation. Sources: Banana, Potato, Pulses.
Vitamin B ₁₂ (Cobalamin)	RBC formation, nerve function. Sources: Meat, Egg, Milk.
Vitamin C (Ascorbic acid)	Immunity booster, helps in healing wounds. Sources: Citrus fruits, Amla, Guava, Tomato.

6. ENZYMES

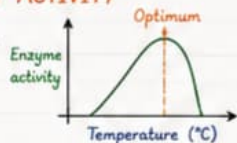
- Enzymes are biological catalysts.
- Mostly proteins in nature.
- Increase the speed of reactions by lowering activation energy.
- Highly specific.
- Not consumed in the reaction.
- Affected by temperature, pH and inhibitors.

ENZYME ACTION – LOCK AND KEY MODEL

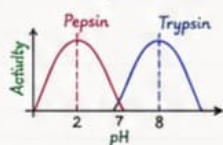


FACTORS AFFECTING ENZYME ACTIVITY

- ★ **Temperature:** Increases activity up to optimum, then decreases (enzymes denature at high temp.).

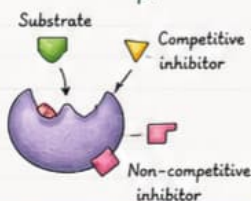


- ★ **pH:** Each enzyme has an optimum pH. Pepsin works in acidic medium (pH ~2) while Trypsin in basic medium (pH ~8).



- ★ **Inhibitors:**

- Competitive – resemble substrate
- Non-competitive – bind at other site and change enzyme shape.



7. HORMONES

- Hormones are chemical messengers produced in endocrine glands.
- Transported by blood to target organs.
- Regulate growth, metabolism, reproduction, etc.

Hormone	Source	Function
Insulin	Pancreas	Lowers blood glucose level.
Glucagon	Pancreas	Raises blood glucose level.
Thyroxine	Thyroid gland	Regulates metabolism, growth & development.
Adrenaline	Adrenal gland	Prepares body for emergency (fight or flight).
Growth Hormone	Pituitary gland	Controls growth of bones and tissues.

IMPORTANCE OF BIOMOLECULES

- ✓ Provide energy and nutrients.
- ✓ Build and repair body tissues.
- ✓ Regulate body processes.
- ✓ Store and transmit genetic information.
- ✓ Protect from diseases and maintain homeostasis.



Quick Recap

- ★ Carbohydrates – Energy
- ★ Proteins – Growth & Repair
- ★ Lipids – Energy Storage
- ★ Nucleic Acids – Genetic Info
- ★ Vitamins – Protective
- ★ Enzymes – Catalysts
- ★ Hormones – Regulators

Key Takeaway.

All biomolecules work together like a team to keep our body running smoothly. A balanced diet and healthy lifestyle are the keys to good health!